

Consecutive Chemical Activation Steps in the OH-Initiated Atmospheric Degradation of Isoprene: An Analysis with Coupled Master Equations

MARK PFEIFLE, MATTHIAS OLZMANN

Institut für Physikalische Chemie, Karlsruher Institut für Technologie, 76131 Karlsruhe, Germany

Received 7 November 2013; revised 18 December 2013; accepted 12 January 2014

DOI 10.1002/kin.20849

Published online 12 February 2014 in Wiley Online Library (wileyonlinelibrary.com).

ABSTRACT: The influence of a two-step chemical activation on 1,5-H and 1,6-H shift reactions of hydroxyl-peroxy radicals formed in the atmospheric photooxidation of isoprene was investigated by means of a master equation analysis. To account for multiple chemical activation processes, three master equations were coupled. The general approach of this coupling is described, and consequences for steady-state regimes are examined. The specific calculations show that chemical activation has no substantial influence on the rate coefficients of the above-mentioned reactions under tropospheric conditions. However, it is demonstrated that high-pressure limits of the thermal rate coefficients instead of the falloff-corrected values have to be used for kinetic modeling. This is a consequence of the continuous population of the high-energy part of the isoprene-OH-O₂ adduct distribution by the forming reactions under steady-state conditions. The rate coefficients of the isomerization reactions at $T = 298\text{ K}$ were calculated to be $k_{3a}^{\infty} = 1.5 \times 10^{-3}\text{ s}^{-1}$ (1,5-H-shift of the 1,2-isomer) and $k_{4a}^{\infty} = 6.5\text{ s}^{-1}$ (1,6-H-shift of the (Z)-1,4-isomer). The calculated value of k_{4a}^{∞} is three orders of magnitude larger than a recently reported experimentally observed rate coefficient for the hydrogen shift reactions of the hydroxyl-peroxy intermediates. It is shown that this discrepancy is in part due to the fact that the experiment does not distinguish between different structural isomers. A comparison of the experimentally determined isotope effect with the calculated value shows a reasonable agreement. © 2014 Wiley Periodicals, Inc. *Int J Chem Kinet* 46: 231–244, 2014

Correspondence to: Matthias Olzmann; e-mail: matthias.olzmann@kit.edu.

Contract grant sponsor: Deutsche Forschungsgemeinschaft.

Contract grant number: OI 85/5-1.

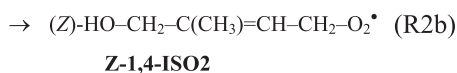
Supporting Information is available in the online issue at www.wileyonlinelibrary.com.

© 2014 Wiley Periodicals, Inc.

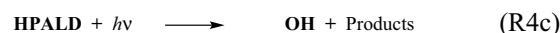
INTRODUCTION

The mechanism of the oxidation of isoprene in the lower troposphere has attracted attention of many atmospheric chemists over the past few years, owing to the large abundance of this biogenic hydrocarbon,

Isoprene is predicted to be a major hydroxyl radical sink due to reaction (R1), followed by the fast consecutive reactions (R2a) and (R2b):



(R4a)


$$\rightarrow \text{HO-C}_5\text{H}_8\text{-O}^\bullet + \text{NO}_2 \quad (\text{R5b})$$

Such chemically activated unimolecular reactions following addition of O₂ to organic radicals have been found in a number of cases: For ethyl [15], acetyl [16,17], and α -carboxyalkyl radicals [14], prompt reactions are effectively suppressed by collisional stabilization at tropospheric temperatures and pressures. They

may become important at higher temperatures or at low pressures. But there are also examples, where prompt isomerizations under tropospheric conditions can occur: In the photochemical oxidation of glyoxal, approximately 80% of the peroxy radicals formed after H abstraction by OH and addition of O₂ react promptly via H shift and subsequent decomposition [18]. Likewise, the α -hydroxyethyl radical, an important intermediate in both atmospheric and combustion chemistry, was shown to react in a chemically activated mechanism after O₂ addition at pressures of up to 50 bar [19]. In the atmospheric degradation of acetylene, the reaction of the chemically activated C₂H₂-OH adduct with O₂ is obviously fast enough to effectively compete with collisional stabilization. Accordingly, a considerable part of the reacting C₂H₂-OH population is vibrationally excited [20].

In a recent theoretical work, Asatryan et al. [21] discussed a *double* activation mechanism in the OH-initiated oxidation of acrolein. Double activation means that excess energy from *two* subsequent addition reactions (with OH and O₂) is available for the unimolecular decomposition of the adduct species. It has to be noted, however, that the argumentation in [21] is based on qualitative features of the potential energy surface (PES) and general considerations on the competition between unimolecular reaction steps and collisional processes. It is not quantitatively supported, for example, by a master equation analysis.

A related result was found by da Silva [22] in a theoretical study on the oxidation of methacrolein. From the solution of a single master equation for the methacrolein + OH + O₂ reaction system, the author concluded that under atmospheric conditions about 50% of the intermediate peroxy radical population bears a significant amount of energy from double activation, which opens up the possibility of prompt unimolecular decomposition reactions.

In the isoprene oxidation mechanism, a similar process is also conceivable. Dibble [23] has investigated the chemically activated *cis-trans* isomerizations of the 1- and 4-isoprene-OH adducts (1-ISOH and 4-ISOH), concluding that the excess energy from the hydroxyl association step leads to a prompt rotation about the central C-C bond. The ISOH intermediates will then either lose their energy by collisions with N₂ or will be captured by O₂. The latter reaction occurs with an overall rate coefficient of $k_{2,\text{total}} = 2.3 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$ [24], where we note that $k_{2,\text{total}}$ includes the reactions to all possible isomers in contrast to k_{2a} and k_{2b} for the isomer-specific reactions above. In air at $P = 1$ bar and $T = 298$ K, this value of $k_{2,\text{total}}$ corresponds to a pseudo-first-order rate coefficient of $1.2 \times 10^7 \text{ s}^{-1}$. Directly after their formation, the re-

sulting peroxy radicals (ISO2) are vibrationally excited due to the exothermicity of the O₂ addition (reaction (R2a) or (R2b)). In addition, there could be residual energy from the OH addition (reaction (R1)). This double activation could then enhance the unimolecular reactions (R3a) and (R4a).

In the current work, we study such multiple activation scenarios by coupling of master equations. By coupling, we mean that the energy distribution of an intermediate from one master equation enters the reactant energy distribution in another one. We study general features of such coupled systems and apply this approach to the atmospheric degradation of isoprene, examining the possibility of OH recirculation in a quantitative way.

METHODOLOGY

Characterization of the Potential Energy Diagram

The potential energy diagram of the overall reaction system is schematically depicted in Fig. 1, and the relevant energy differences and their sources are collected in Table I. In the present work, only the reaction sequence initiated by OH addition to the 1-position of isoprene, which is kinetically favored with a branching ratio of ca. 60% [4,25] is considered. In this way, all the essential kinetic features are captured, while unnecessary complexity due to different isomers is avoided. For the second step, the O₂ addition, the two ISO2 isomers accessible from 1-ISOH, which can undergo 1,5-H shift (1,2-ISO2) and 1,6-H shift (Z-1,4-ISO2), are taken into account.

Threshold energies and energies of reaction were taken from the literature, except $\Delta_R E_{4a}$, which was calculated from density functional theory at B3LYP/6-311++G(d,p) level of theory [26]. For the energies of reaction of (R1) and (R2a) as well as for the energy barrier of the internal rotation of 1-ISOH, values from different quantum chemical methods were available and we used simple averages for our calculations (see Table I). Of course, such an averaging is not per se meaningful, but it turns out that the general kinetic behavior of the overall reaction system is not influenced by the specific values of these quantities. This is discussed in more detail below.

The Master Equation

The general scheme for a chemically activated reaction with a consecutive bimolecular step of the intermediate C can be expressed in the following way:

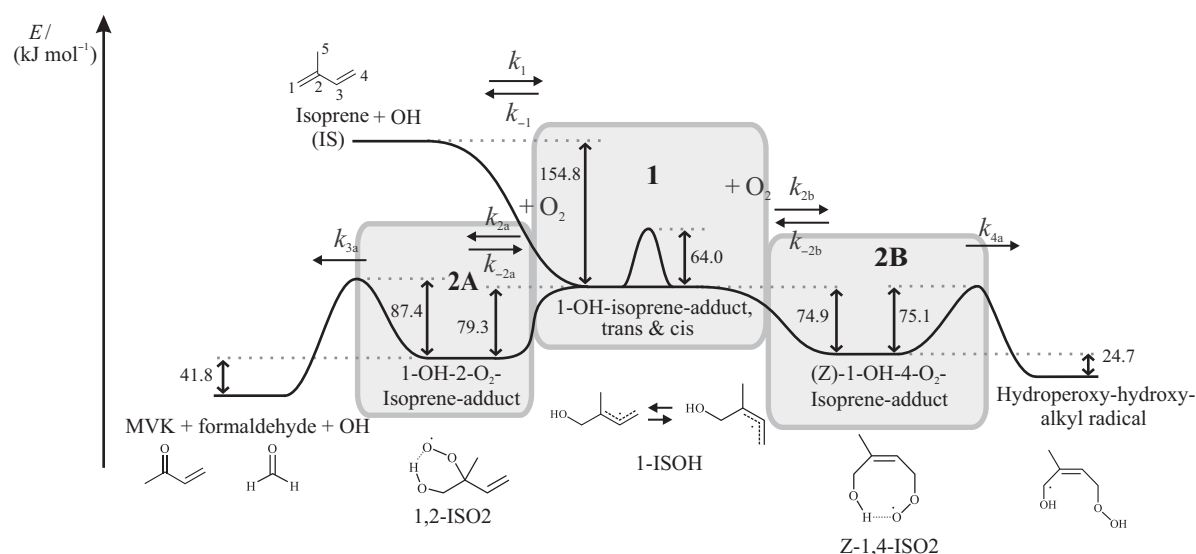


Figure 1 Schematic energy diagram (cf. Table I).

Table I Energy Differences (Including Zero-Point Energies) Used in Our Calculations (cf. Fig. 1)

	Value (kJ mol ⁻¹)	Reference
$\Delta_R E_1$	-154.8	[4,5,59–62] ^a
$\Delta^\ddagger E_{c/t}$	64.0	[5,23] ^a
$\Delta_R E_{2a}$	-79.3	[4,5,59] ^a
$\Delta_R E_{2b}$	-74.9	[5]
$\Delta^\ddagger E_{3a}$	87.4	[5]
$\Delta^\ddagger E_{4a}$	75.1	[6]
$\Delta_R E_{3a}$	-41.8	[7]
$\Delta_R E_{4a}$	-24.7	This work ^b

^a Average (see the text).

^b Calculated with density functional theory at B3LYP/6-311++G(d,p) level of theory.



Here, the symbol * indicates that the intermediate C is chemically activated, i.e., obeys a nonthermal energetic distribution, which is influenced by reactions (a)–(c) and collisions with the bath gas. For general aspects of chemical activation, the reader is referred to the monographs [11–13] and for the influence of consecutive bimolecular steps to [17,27–30].

This distribution of C, $n(E, t)$, can be obtained by solving the following master equation [11–13]:

$$\begin{aligned} \frac{dn(E, t)}{dt} = & R_a f(E) - \omega n(E, t) \\ & + \omega \int_0^\infty P(E, \varepsilon) n(\varepsilon, t) d\varepsilon \\ & - \{k_{-a}(E) + k_b(E) + k_c[D]\} n(E, t) \quad (1) \end{aligned}$$

Here, R_a is the rate of reaction of step (a), $f(E)$ denotes the normalized nascent distribution of C as produced in step (a), ω is the collision frequency, $P(E, \varepsilon)$ is the collisional transition probability for a transition $\varepsilon \rightarrow E$, $k_i(E)$ is the energy-specific rate coefficient of the unimolecular reaction (i), k_c denotes the rate coefficient of the bimolecular reaction (c), and $[D]$ is the concentration of D, where we assume pseudo-first-order conditions, that is, a large excess of D. We also note that k_c principally depends on the excitation energy E , but because little is known about this energy dependence for large molecules, the thermal rate coefficient is used. As is customary, the rovibrational ground state of the intermediate C is taken as the energy zero of the master equation.

Equation (1) can be discretized in energy bins of size ΔE and cast into a matrix form [11–13]:

$$\begin{aligned} (d/dt)\mathbf{N}(t) = & R_a \mathbf{F} - [\omega(\mathbf{I} - \mathbf{P}) + \mathbf{K} + k_c[D]\mathbf{I}] \mathbf{N}(t) \\ \equiv & R_a \mathbf{F} - \mathbf{JN}(t) \quad (2) \end{aligned}$$

Here, $\mathbf{I}$ is the unit matrix, and the other vector/matrix symbols correspond to those of Eq. (1). The solution

for the initial condition $\mathbf{N}(t = 0) = \mathbf{0}$ and a constant input flux $R_a \mathbf{F}$ switched on at $t = 0$ reads as [31–33]

$$\mathbf{N}(t) = R_a \sum_{i=1}^n e_i \frac{1 - \exp(-\lambda_i t)}{\lambda_i} \mathbf{E}_i \quad (3)$$

where λ_i and $\mathbf{E}_i$ are the eigenvalues and eigenvectors, respectively, of the matrix $\mathbf{J}$, which is defined in Eq. (2). The expansion coefficients e_i are obtained from the projection of $\mathbf{F}$ onto the eigenvectors $\mathbf{E}_i$:

$$\mathbf{F} = \sum_{i=1}^n e_i \mathbf{E}_i \quad (4)$$

Under steady-state conditions, $d\mathbf{N}/dt = \mathbf{0}$; from Eq. (2) it follows $R_a \mathbf{F} = \mathbf{J} \mathbf{N}^{\text{ss}}$, and one obtains the steady-state solution:

$$\mathbf{N}^{\text{ss}} = R_a \mathbf{J}^{-1} \mathbf{F} \quad (5)$$

which can be normalized if required.

The observable rate coefficient of the chemically activated unimolecular reaction j , $k_j^{(\text{ca})}(T, P)$, can then be calculated by averaging the specific rate coefficients $k_j(E)$ over the normalized (symbol $\sim$) steady-state distribution $\tilde{n}^{\text{ss}}(E)$:

$$k_j^{(\text{ca})}(T, P) = \int_0^\infty k_j(E) \tilde{n}^{\text{ss}}(E; T, P) dE \quad (6)$$

We note that the rate coefficient for the *thermal* unimolecular reaction of C, $k_j^{(\text{th})}(T, P)$, can also be obtained from Eq. (6), if the *thermal* steady-state distribution is used. The sum of these rate coefficients $k_j^{(\text{th})}(T, P)$ over all channels j is identical to the lowest eigenvalue of the matrix $\mathbf{J}$ [11–13].

The nascent distribution of C resulting from a bimolecular combination reaction of reactants, which are in thermal equilibrium, is obtained from [11–13]

$$f(E) = \frac{W_{-a}(E - E_{0(-a)}) \exp[-(E - E_{0(-a)})/k_B T]}{\int_0^\infty W_{-a}(\varepsilon) \exp[-\varepsilon/k_B T] d\varepsilon} \quad (7)$$

with $E \geq E_{0(-a)}$, where $E_{0(-a)}$ denotes the threshold energy of the reverse reaction of the formation step (a), and W_{-a} represents the sum of states of its transition state; k_B is Boltzmann's constant.

In case of nonthermal reactants A and B with normalized populations $\tilde{n}_A(\varepsilon)$ and $\tilde{n}_B(\varepsilon)$, respectively, the

nascent distribution of C can be approximated by the shifted convolution [27]

$$f(E) = \int_0^{E-E_{0(-a)}} \tilde{n}_A(\varepsilon) \tilde{n}_B(E - E_{0(-a)} - \varepsilon) d\varepsilon \quad (8)$$

with $E \geq E_{0(-a)}$. Here, it is assumed that the reactive cross section of reaction (a) is independent of the internal energies of the reactants A and B, which is equivalent to the assumption that the rate of the reverse reaction (–a) can be described by (vibrational) phase space theory neglecting angular momentum restrictions (see discussions in [34]).

The specific rate coefficients $k_j(E)$ were obtained from statistical rate theory. Their calculation is separately described in the next section.

For our calculations, a master equation was set up with a uniform bin size of $\Delta E = 10 \text{ cm}^{-1}$. The stepladder model obeying detailed balancing was used for the collisional transition probabilities [11,12], where the step size, ΔE_{SL} , which corresponds to the average energy transferred per down collision, was treated as a variable parameter. We performed calculations for $\Delta E_{\text{SL}} = 50, 100$, and 200 cm^{-1} , which should cover a reasonable range for collisions of medium-sized oxygenated hydrocarbons with N_2 and O_2 [11,35]. For the calculation of the collision frequency ω , the Lennard–Jones parameters of 2-methyl-1-butanol as a proxy for the intermediate species of isoprene degradation were used. From the critical data of 2-methyl-1-butanol [36], we obtained $\varepsilon/k_B = 520 \text{ K}$ and $\sigma = 5.42 \text{ \AA}$, using the empirical method of Stiel and Thodos [37]. Combined with the Lennard–Jones parameters for N_2 and O_2 [38], this yields a collision frequency of $\omega \sim 1.2 \times 10^{10} \text{ s}^{-1} \text{ bar}^{-1} \times P$ for both the isoprene-OH and the isoprene-OH- O_2 adduct in air at $T = 298 \text{ K}$ (within two-digit accuracy).

Because of the stepladder model for the collisional transition probabilities, the matrix $\mathbf{J}$ is tridiagonal. After symmetrization [11], its eigenvalues and eigenvectors were calculated by using the EISPACK routine tq12 [39] (see also [40]). The steady-state solution, $\mathbf{N}^{\text{ss}}$, of Eq. (2) is most conveniently obtained by directly solving $R_a \mathbf{F} = \mathbf{J} \mathbf{N}^{\text{ss}}$ for $\mathbf{N}^{\text{ss}}$ (routines banbks and bandec from [40]) and subsequent normalization. For more technical details of our approach, see [28,29,41].

Calculation of Specific Rate Coefficients

The specific rate coefficients of the unimolecular reactions steps were obtained from statistical rate

theory [11–13]:

$$k_j(E) = \frac{W_j(E - E_{0(j)})}{h\rho(E)} \quad (9)$$

where $W_j(E - E_{0(j)})$ is the cumulative reaction probability of reaction j and $\rho(E)$ is the density of states of the reactant; h is Planck's constant. Both W_j and ρ were determined for a total angular momentum quantum number $J = 42$ with exact counting routines [11–13,42]. This value of J corresponds to the thermally averaged rotational excitation of isoprene at $T = 298$ K. To assess the influence of angular momentum contributions from addition of OH and O₂ [43], we also tested $J = 42 \pm 20$ but found no significant effects on the calculated overall rate coefficients and branching ratios. The threshold or dissociation energies $E_{0(j)}$ were adopted from published high-level quantum chemical calculations and are listed in Table I along with the corresponding references. Harmonic vibrational wave numbers, rotational constants, and torsional potential curves were calculated from density functional theory at B3LYP/6–311++G(d,p) level of theory with the Gaussian 03 program package [26]. The results are compiled in the Supporting Information.

For the specific rate coefficients of the 1,5-H shift (reaction (R3a)) and 1,6-H shift (reaction (R4a)), which are both characterized by pronounced barriers along their respective reaction coordinates, the Rice–Ramsperger–Kassel–Marcus (RRKM) theory [11–13, 44,45] was used. To account for tunneling, a one-dimensional correction [46,47] based on the semi-classical Wentzel–Kramers–Brillouin approximation [48–50] was employed. The transmission coefficient was calculated on the basis of a cubic spline interpolation [40] of the intrinsic reaction coordinate (IRC) curves reported by Peeters et al. [5,6]. These were constructed from B3LYP/6–31+G(d,p) geometries and CBS-QB3 single-point energies.

For reactions (R-1), (R-2a), and (R-2b), which are dissociation steps without a distinct potential energy barrier for the reverse combination reaction, the simplified statistical adiabatic channel model (SACM) [11,12,42,51,52] was used to calculate $k_j(E_i)$. The SACM interpolation parameters, α/β , were fitted to the thermal recombination rate coefficients at 298 K [51]. The following rate coefficients were used [5]: $k_1(298 \text{ K}) = 6.0 \times 10^{-11} \text{ cm}^3 \text{ s}^{-1}$ (from the recommended rate coefficient $k_{1,\text{total}} = 1.0 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ for the overall OH addition [53] and a relative branching fraction of 60% [4,25] for 1-ISOH), $k_{2a}(298 \text{ K}) = 1.5 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$, and $k_{2b}(298 \text{ K}) = 7 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$. From this approach, we obtained $\alpha/\beta = 0.28$ for k_{-1} , $\alpha/\beta = 0.30$ for k_{-2a} , and $\alpha/\beta = 0.28$ for k_{-2b} . We note

that the *simplified* SACM used here is essentially equivalent to other two-parameter formulations of statistical rate theory with an entropic and energetic parameter [11]. These are represented here by the interpolation ratio α/β and the bond-dissociation energy $E_0(J = 0)$, respectively. A parameter-free, detailed SACM calculation on the basis of exact channel eigenvalues is currently far from being feasible for molecules with this degree of complexity.

For both the RRKM and the SACM calculations, the internal degrees of freedom were treated as harmonic oscillators (HO) with wave numbers scaled by 0.97 [54]. Exceptions were made for torsional vibrations about C–O and C–C bonds with bond orders <2. These were treated in analogy to [55] in a one-dimensional hindered internal rotor (HIR) approach on the basis of relaxed PES scans. In the RRKM calculations, such HIR corrections were applied to both $\rho(E)$ and $W_j(\varepsilon)$. In the SACM calculations, these corrections can be directly applied only to $\rho(E)$, because reliable hindrance potentials in the transition range between the reactant and products are not available. Instead, the number of open reaction channels, $W_j(\varepsilon)$, was first calculated by using the well-known simplified interpolation scheme [42,52] and then multiplied by a correction factor $f_{\text{corr}}(\varepsilon) = W_{\text{frag,HIR}}(\varepsilon)/W_{\text{frag,HO}}(\varepsilon)$. This correction factor was determined from the sum of states of the separated fragments in the HIR approximation and HO approximation, respectively. That is, the influence of hindered internal rotors on the number of open reaction channels is approximated by the effect of hindered internal rotors on the sum of states of the separated fragments. This seems appropriate for the low temperatures under atmospheric conditions, that is, low excess energies, where the channel maxima are located close to the separated fragments.

Setup and Coupling of Master Equations

For each well displayed in Fig. 1 (**1**, **2A**, and **2B**), a master equation was set up and solved. Well **1** corresponds to 1-ISOH, which was treated as a single species with a hindered internal rotation about the C2–C3 axis. This rotation represents the energetically accessible cis–trans isomerization with a hindrance potential of 64.0 kJ mol^{−1} (cf. Fig. 1). Besides the entrance channel (R1) and its unimolecular backward reaction (R-1), the bimolecular O₂ addition channels (R2a) and (R2b) were included in the master equation by a total pseudo–first-order rate coefficient ($k_{2,\text{pseudo}} = k_{2a}[\text{O}_2] + k_{2b}[\text{O}_2]$, where a third channel contributing less than 10% [5] was neglected). The nascent energy distribution, $f_1(E)$, was calculated from Eq. (7) with $W_{-1}(\varepsilon)$ from SACM.

Wells **2A** and **2B** correspond to 1,2-ISO2 and Z-1,4-ISO2, respectively. In both cases, besides the formation reactions ((R2a) and (R2b), respectively), their unimolecular back reactions ((R-2a) and (R-2b), respectively) and the isomerization reactions ((R3a) and (R4a), respectively) were included in the master equations. We note that (R3a) is the rate-determining step followed by a fast decomposition of the isomerization product to yield methyl vinyl ketone (MVK) + formaldehyde + OH [5,7] (cf. Fig. 1).

Unlike 1-ISOH in well **1**, the intermediates 1,2-ISO2 in well **2A** and Z-1,4-ISO2 in well **2B** are formed in bimolecular steps with one reactant (1-ISOH) in a nonthermal, chemically activated energy distribution. Accordingly, the nascent energy distributions of 1,2-ISO2, $f_{2A}(E)$, and Z-1,4-ISO2, $f_{2B}(E)$, have to be calculated from Eq. (8). That is, the output of the first master equation governs the input of a second one. This is what we mean by coupling of master equations. With a thermal distribution for O_2 , $\tilde{n}_{O_2}(E)$, and the normalized steady-state distribution for 1-ISOH, $\tilde{n}_1^{ss}(E)$, from the master equation for well **1**, it follows that

$$f_{2A}(E) = \int_0^{E+\Delta_R E_{2a}} \tilde{n}_{O_2}(\varepsilon) \tilde{n}_1^{ss}(E + \Delta_R E_{2a} - \varepsilon) d\varepsilon \quad (10)$$

and

$$f_{2B}(E) = \int_0^{E+\Delta_R E_{2b}} \tilde{n}_{O_2}(\varepsilon) \tilde{n}_1^{ss}(E + \Delta_R E_{2b} - \varepsilon) d\varepsilon \quad (11)$$

If the width of $\tilde{n}_{O_2}(\varepsilon)$ is neglected compared to the width of $\tilde{n}_1^{ss}(\varepsilon)$ (see below), one can set $\tilde{n}_{O_2}(\varepsilon) = \delta(\varepsilon - \langle E \rangle_{O_2})$, where δ denotes the Dirac delta function and $\langle E \rangle_{O_2}$ is the average thermal energy of O_2 . It follows that

$$f_{2A}(E) = \tilde{n}_1^{ss}(E + \Delta_R E_{2a} - \langle E \rangle_{O_2}) \quad (12)$$

and

$$f_{2B}(E) = \tilde{n}_1^{ss}(E + \Delta_R E_{2b} - \langle E \rangle_{O_2}) \quad (13)$$

That is, the nascent distribution of the corresponding intermediate is approximated by the distribution of the chemically activated reactant shifted to higher energies by the sum of the reaction energy at 0 K and the average thermal energy of O_2 [27]. Recall that the energy zero of the master equation is the rovibrational ground state of the intermediate considered.

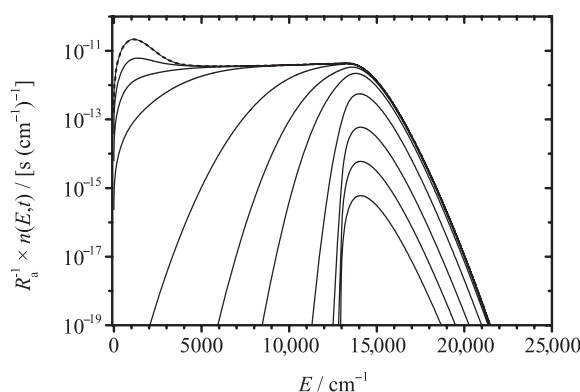


Figure 2 Chemical activation populations of 1-ISOH (well **1**) for $T = 298$ K, $P = 1$ bar, and $\Delta E_{SL} = 100$ cm^{-1} ; (solid lines from bottom to top) $n(E, t)$ from Eq. (3) for $t = 10^{-12}$, 10^{-11} , 10^{-10} , 10^{-9} , 5×10^{-9} , 10^{-8} , 2×10^{-8} , 5×10^{-8} , 7×10^{-8} , 10^{-7} , and 10^{-6} s; (dashed line) $n^{ss}(E)$ from Eq. (5); note that $n(E, t = 10^{-6}$ s) and $n^{ss}(E)$ virtually coincide (for explanations, see text).

Time-Dependent Solutions

Before discussing steady-state solutions of coupled master equations, it is instructive to consider time-dependent solutions of uncoupled master equations and their relation to steady states.

In Fig. 2, the time-dependent molecular populations of 1-ISOH as obtained from Eq. (3) are shown. The initial condition is $n(E, t = 0) = 0 \forall E$. At $t = 0$, a constant flux $R_a f_1(E)$ (with a minimum energy at $E = -\Delta_R E_1 \sim 12,940$ cm^{-1} , cf. Table I) is switched on. The population then increases until after about 10^{-6} s, the steady-state population $n^{ss}(E)$ is reached. Note that this steady-state population is maintained by production of 1-ISOH in reaction (R1), back dissociation in reaction (R-1), and bimolecular capture by O_2 in reactions (R2a) and (R2b). It is these bimolecular capture reactions that prevent the molecular population from becoming thermalized even in the long-time limit. The nearly constant steady-state population at energies between 4000 and 13,000 cm^{-1} is a manifestation of the collisional deactivation cascade between the energy range of the incoming flux above 12,940 cm^{-1} and the energy range of the thermal distribution indicated by the hump at energies below 3000 cm^{-1} .

The situation is different, if no bimolecular capture reaction occurs. The consequences are exemplarily shown in Fig. 3 for the population of 1,2-ISO2 produced in reaction (R2a) from *thermal* 1-ISOH and thermal O_2 . A constant flux obeying a shifted Boltzmann distribution (calculated from Eq. (7)) with a minimum energy at $E = -\Delta_R E_{2a} \sim 6630$ cm^{-1} (cf. Table I) is switched on at $t = 0$. The population grows

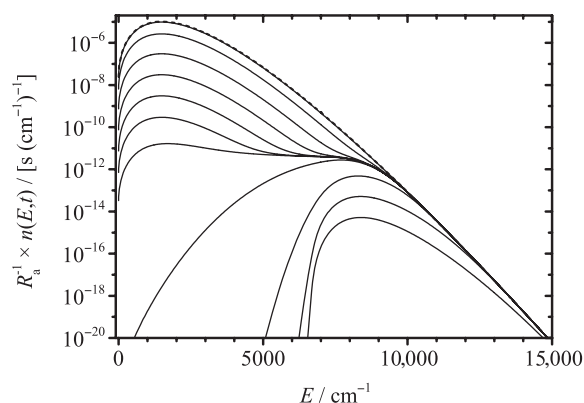


Figure 3 Chemical activation populations of 1,2-ISO2 (well **2A**) formed from thermal 1-ISOH for $T = 298$ K, $P = 1$ bar, and $\Delta E_{\text{SL}} = 100$ cm^{-1} ; (solid lines from bottom to top) $n(E, t)$ from Eq. (3) for $t = 10^{-11}$, 10^{-10} , 10^{-9} , 10^{-8} , 10^{-7} , 10^{-6} , 10^{-5} , 10^{-4} , 10^{-3} , 10^{-2} , and 10^{-1} s; (dashed line) $n^{\text{ss}}(E)$ from Eq. (5); note that $n(E, t = 10^{-1}$ s) and $n^{\text{ss}}(E)$ virtually coincide (for explanations, see the text).

with time and reaches steady state after about 10^{-1} s. In this case, the steady-state population obeys a stabilized, Boltzmann-like distribution in contrast to the rather flat steady-state population shown in Fig. 2. The situation for Z-1,4-ISO2 is analogous.

Steady-State Solutions

In general, for chemically activated intermediates two different steady-state regimes can be distinguished [31,32]:

- at earlier times, the so-called intermediate steady state, where only the concentration at high energies is constant, and the population at lower energies steadily grows (cf. Fig. 3, $t \geq 10^{-7}$ s),
- at longer times, the final steady-state, where also the low-energy states are filled up, and the total concentration of the intermediate remains constant (cf. Fig. 3, $t \geq 10^{-1}$ s).

The transition between the two regimes occurs at times $t \sim 0.1 \times \tau_{\text{therm}}$ [31] where τ_{therm} denotes the thermal lifetime of the intermediate at the given temperature and pressure. For 1,2-ISO2 and Z-1,4-ISO2, we calculated thermal lifetimes between 5 and 50 ms under atmospheric conditions (see the Results and Discussion section). Accordingly, we assume final steady state for these species, because also the concentrations of the reactants forming them (IS, OH, and O_2) do not significantly change on this timescale under atmospheric conditions.

From the discussions of the preceding section, it becomes obvious that the final steady state becomes identical to the intermediate steady state, if a sufficiently fast bimolecular reaction occurs, that removes the stabilized intermediates [28]. By such a reaction, the population at low energies is prevented from being built up. The general behavior can be realized from Fig. 2, where $n^{\text{ss}}(E)$ is already reached after 10^{-6} s (compared to 10^{-1} s in the case of 1,2-ISO2, cf. Fig. 3), and its shape is very similar to the shape of a typical intermediate steady-state population such as $n(E, t = 10^{-7}$ s) in Fig. 3.

RESULTS AND DISCUSSION

1,2-ISO2 and Z-1,4-ISO2 from Double Activation

In Fig. 4, the steady-state distributions for well **1** are shown. For comparison, the nascent and high-pressure distributions are also displayed. Note that the curves for the specific rate coefficient of the unimolecular back dissociation (R-1) and the nascent distribution of the intermediate 1-ISOH necessarily start at the same energy, namely $E_{0(-)} = -\Delta_R E_1 = 154.8$ $\text{kJ mol}^{-1} = 12,940$ cm^{-1} (cf. Table I). It can be realized that 1-ISOH does not reach thermal equilibrium under atmospheric conditions. The rapid bimolecular reaction with O_2 , which is most competitive for the stabilized that is, long-lived 1-ISOH species, causes the thermal and the steady-state distribution to deviate from each other. Under steady-state conditions, there is a notable population also at higher energies up to 13,000 cm^{-1} , which is maintained by the incoming flux from

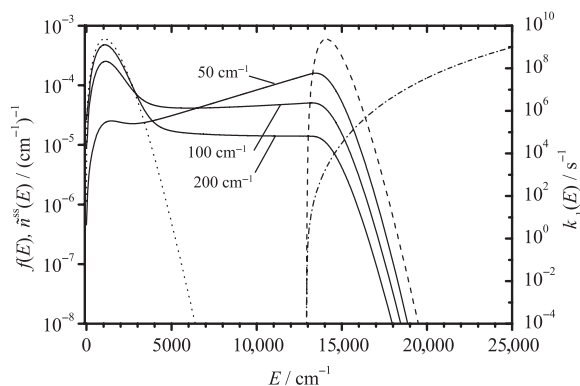


Figure 4 Normalized steady-state energy distributions of 1-ISOH (well **1**) for $T = 298$ K and specific rate coefficients; (dashed line) nascent distribution $f(E)$; (solid lines) steady-state distributions $\tilde{n}^{\text{ss}}(E)$ for different values of ΔE_{SL} and $P = 1$ bar; (dotted line) Boltzmann distribution; (dot-dashed line) specific rate coefficient $k_{-1}(E)$.

Table II Calculated Rate Coefficients for the Unimolecular Steps of 1-ISOH, 1,2-ISO2, and Z-1,4-ISO2

ΔE_{SL}	$k_i^{(ca)}(298\text{ K}, 1\text{ bar}) (\text{s}^{-1})$			$k_i^{(th)}(298\text{ K}, 1\text{ bar}) (\text{s}^{-1})$		
	50 cm^{-1}	100 cm^{-1}	200 cm^{-1}	50 cm^{-1}	100 cm^{-1}	200 cm^{-1}
k_{-1}	6.4×10^3	1.9×10^3	5.1×10^2	2.3×10^{-13}	2.3×10^{-13}	2.3×10^{-13}
k_{-2a}	6.6×10^2	8.1×10^1	5.1×10^1	2.1×10^1	3.1×10^1	4.0×10^1
k_{3a}	2.8×10^{-2}	2.8×10^{-3}	1.7×10^{-3}	7.5×10^{-4}	1.0×10^{-3}	1.3×10^{-3}
k_{-2b}	5.7×10^3	6.5×10^2	4.3×10^2	9.4×10^1	1.7×10^2	2.7×10^2
k_{4a}	7.9×10^0	6.3×10^0	6.3×10^0	5.4×10^0	5.8×10^0	6.1×10^0
k_{-2a}/k_{3a}	2.4×10^4	2.9×10^4	3.1×10^4	2.8×10^4	3.1×10^4	3.2×10^4
k_{-2b}/k_{4a}	7.2×10^2	1.0×10^2	6.7×10^1	1.8×10^1	3.0×10^1	4.4×10^1

$k_i^{(ca)}$ denotes chemical activation, and $k_i^{(th)}$ stands for thermal activation.

reaction (R1). But with increasing ΔE_{SL} , also the stabilized fraction increases.

In Table II, rate coefficients calculated from Eq. (6) for $T = 298\text{ K}$ and $P = 1\text{ bar}$ are collected. For comparison, chemical activation ($k_i^{(ca)}$) and thermal activation ($k_i^{(th)}$) are considered. It is evident that the rate coefficient for prompt back dissociation, k_{-1} , is increased by 16 orders of magnitude compared to its thermal counterpart, $k_{-1}^{(th)}$. Nonetheless, back dissociation can still not compete with O_2 addition occurring with $k_{2,\text{total}} = 1.2 \times 10^7\text{ s}^{-1}$ (see the Introduction). Consequently, the steady-state concentration of 1-ISOH is virtually the same regardless of whether the rate coefficient k_{-1} for thermal or chemical activation is used. Under atmospheric conditions, this concentration is governed by k_1 and $k_{2,\text{total}}$ and much less by k_{-1} .

Figure 5 illustrates how the steady-state energy distributions of 1,2-ISO2 and Z-1,4-ISO2 are affected by the nonequilibrium distribution of 1-ISOH. The energy distribution of 1-ISOH determines the input distributions of 1,2-ISO2 and Z-1,4-ISO2 via Eqs. (12) and (13). This can be realized by comparing the curves for $f(E)$ from Fig. 5 (upper panel) with those for $\tilde{n}^{ss}(E)$ from Fig. 4. The differences only consist of a shift along the energy scale by $-\Delta_R E_{2a} + \langle E \rangle_{\text{O}_2}$ (cf. Eq. (12)). In contrast to shifted Boltzmann-like distributions for chemical activation from thermal reactants, as, for example, $f(E)$ in Fig. 4, the distributions arising from chemically activated reactants are rather broad. As a consequence, the steady-state distributions $\tilde{n}^{ss}(E)$ of 1,2-ISO2 and Z-1,4-ISO2 exhibit small residual parts reaching up to high energies around $20,000\text{ cm}^{-1}$. These parts correspond to “double activation” but represent only a vanishingly small fraction of the total population. Apart from this feature, the $\tilde{n}^{ss}(E)$ essentially coincides with the Boltzmann distribution.

This is also reflected in the calculated values of $k_i^{(ca)}$, which are listed in Table II. They are generally larger

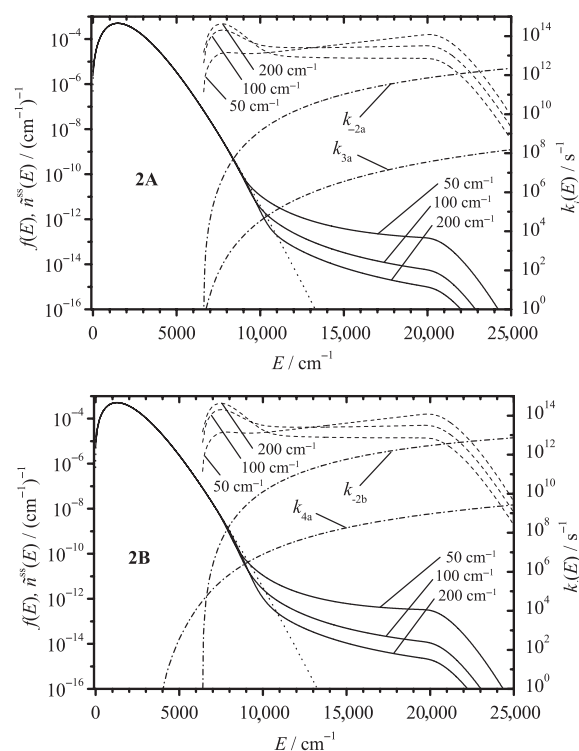


Figure 5 Normalized steady-state energy distributions and specific rate coefficients of 1,2-ISO2 (well 2A, upper panel) and Z-1,4-ISO2 (well 2B, lower panel) for $T = 298\text{ K}$ and $P = 1\text{ bar}$; (dashed lines) nascent distributions $f(E)$ for different values of ΔE_{SL} ; (solid lines) steady-state distributions $\tilde{n}^{ss}(E)$ for different values of ΔE_{SL} ; (dotted line) Boltzmann distribution; (dot-dashed lines) specific rate coefficient $k_i(E)$.

than the corresponding values of $k_i^{(th)}$. For large energy transfer parameters ($\Delta E_{SL} = 100$ and 200 cm^{-1}), both rate coefficients are in the same order of magnitude. For $\Delta E_{SL} = 50\text{ cm}^{-1}$, the influence of chemical activation is more pronounced, leading to an increase by 1–2 orders of magnitude. Here, the smaller collision efficiency

Table III Calculated Rate Coefficients for the Unimolecular Steps of 1,2-ISO2 and Z-1,4-ISO2 Assuming Single Chemical Activation from 1-ISOH + O₂

ΔE_{SL}	$k_i^{(ca,single)}(298\text{ K}, 1\text{ bar}) (\text{s}^{-1})$			$k_i^\infty(298\text{ K}) (\text{s}^{-1})$
	50 cm ⁻¹	100 cm ⁻¹	200 cm ⁻¹	
k_{-2a}	4.9×10^1	4.9×10^1	4.9×10^1	5.0×10^1
k_{3a}	1.5×10^{-3}	1.5×10^{-3}	1.5×10^{-3}	1.5×10^{-3}
k_{-2b}	5.3×10^2	5.2×10^2	5.1×10^2	5.2×10^2
k_{4a}	6.6×10^0	6.5×10^0	6.5×10^0	6.5×10^0
k_{-2a}/k_{3a}	3.2×10^4	3.2×10^4	3.2×10^4	3.3×10^4
k_{-2b}/k_{4a}	8.1×10^1	7.9×10^1	7.9×10^1	8.0×10^1

causes a higher fraction of energized molecules, which react with larger specific rate coefficients.

In the last two lines of Table II, the ratios k_{-2a}/k_{3a} and k_{-2b}/k_{4a} are given. It is evident that the backward reactions dominate the isomerization reactions in all cases. Therefore, the inclusion of chemical activation does not alter the situation already described in [5] for the case of thermal reactions: The O₂ addition–dissociation reaction is nearly in a chemical equilibrium, which is only slightly interfered by the consecutive reactions. As a consequence, the abundances of the different ISO2 isomers derived from a given ISOH isomer depend on their relative *thermodynamic* stability, because interconversion is possible via repeated O₂ elimination and readdition.

1,2-ISO2 and Z-1,4-ISO2 from Single Activation

As mentioned in the Introduction, da Silva et al. [7,14] argued that repeated back dissociation of O₂ from ISO2 species and reassociation could enhance the yield of the isomerization reactions via chemical activation effects. However, it is obvious that the excess energy of 1-ISOH due to reaction (R1) is completely lost after several dissociation–addition cycles (R-2a)–(R2a) and (R-2b)–(R2b). This is not a result from our master equation calculations but can be rationalized from the following general considerations: The total pseudo–first-order rate coefficient for addition at $T = 298\text{ K}$ and $P = 1\text{ bar}$ is $k_{2,\text{total}}[\text{O}_2] \sim 1.2 \times 10^7\text{ s}^{-1}$ (see the Introduction), whereas the collision frequency at the same conditions is in the order of 10^{10} s^{-1} . It follows that after a short induction period the majority of ISOH radicals present at a given time does not directly originate from reaction (R1), but has already gone through many O₂ addition–dissociation cycles. Therefore, the ISOH radicals will be essentially characterized by a Boltzmann distribution. This means that a double activation mechanism is not adequate. Instead a single-

activation mechanism has to be used to calculate k_{-2a} , k_{3a} , k_{-2b} , and k_{4a} . The master equations have to be set up accordingly. The nascent distributions of 1,2-ISO2 and Z-1,4-ISO2 then stem from equilibrium populations of 1-ISOH and O₂, so they can be calculated with Eq. (7). This also eliminates the influence of the actual value of $\Delta_R E_1$ on the kinetics of the consecutive steps.

Table III lists the results for these conditions. It can be seen that the rate coefficients of the unimolecular reactions of 1,2-ISO2 and Z-1,4-ISO2 are virtually independent of ΔE_{SL} and equal to their high-pressure limiting values. This means that collisional stabilization is fast enough to dissipate the exoergicity of the addition step before any prompt unimolecular reactions can take place. To examine whether this behavior is sensitive to the reaction energies $\Delta_R E_{2a}$ and $\Delta_R E_{2b}$, we varied these quantities by $\pm 700\text{ cm}^{-1}$, leaving the threshold energies for the isomerization reactions, $\Delta^\ddagger E_{3a}$ and $\Delta^\ddagger E_{4a}$, constant. Within this parameter range, no effect on the energetic distributions and on the rate coefficients k_{3a} and k_{4a} was observed.

Accordingly, the high-pressure limits of the rate coefficients govern the kinetics of the system. It is interesting to note that a *thermal* master equation approach does not yield the same results. Our corresponding calculations revealed that all *thermal* unimolecular reactions of 1,2-ISO2 and Z-1,4-ISO2 are still in the falloff regime at 1 bar, albeit not far away from the high-pressure limit. However, such an approach neglects the fact that the forming reactions (R2a) and (R2b) constantly populate the high-energy levels of the chemically activated intermediate and in this way compensate for the population depletion caused by the unimolecular reactions. A similar behavior was already discussed in [29] for the intermediate HSO₅ radical in the atmospheric SO₂ oxidation mechanism. That is, even though our study shows that chemical activation does not affect the branching ratios in the considered system, it is clear that a thermal master equation would

give wrong results for the rate coefficients of the specific steps.

Comparison with Other Theoretical Works

The Arrhenius expressions reported by Peeters et al. [5,6] yield $k_{3a} = 4 \times 10^{-3} \text{ s}^{-1}$ and $k_{4a} = 0.7 \text{ s}^{-1}$ at 298 K. It should be noted that these authors considered their data as lower limits. The agreement between those values and the rate coefficients calculated in this work ($k_{3a} = 1.5 \times 10^{-3} \text{ s}^{-1}$ and $k_{4a} = 6.5 \text{ s}^{-1}$) appears reasonable. This is expected, because the energy barriers we used were adopted from their publication [5]. Deviations can be explained by the partially different treatment of internal degrees of freedom and conformers. Whereas Peeters et al. used a multiconformer transition state theory approach with total partition functions calculated by a weighted summation of single-conformer partition functions, we accounted for different conformers by the one-dimensional hindered internal rotor approach outlined above.

The rate coefficient of $k_{3a}(298 \text{ K}) = 5 \times 10^{-4} \text{ s}^{-1}$ for the 1,5-H shift reported by da Silva et al. [7] is considerably lower than our value. This is probably due to a 1.7 kJ mol^{-1} higher energy barrier and a different treatment of tunneling in their work.

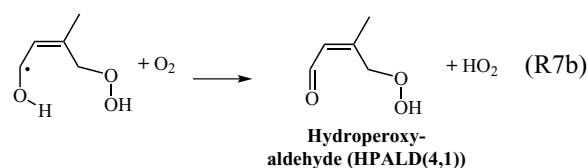
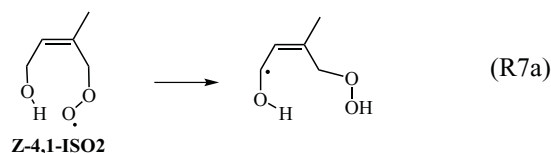
Comparison with Recent Experimental Findings

The rate coefficient, k_{isom} , for the production of hydroperoxyaldehydes (HPALD) via the (rate-determining) isomerizations of the ISO2 intermediates and subsequent reaction with O_2 was recently determined by Crounse et al. [56]. The authors performed a relative rate study in a smog chamber with mass-spectrometric detection. The reference reaction was



where ISOPOOH stands for the hydroxyl hydroperoxides formed from ISO2 [56]. At $T = 295 \text{ K}$, the authors obtained a rate coefficient $k_{\text{isom}} = 2 \times 10^{-3} \text{ s}^{-1}$, and their Arrhenius expression yields $k_{\text{isom}} = 2.5 \times 10^{-3} \text{ s}^{-1}$ for $T = 298 \text{ K}$. This is three orders of magnitude smaller than the values for k_{4a} predicted in the current study and in the studies by Peeters et al. [5,6]. However, as the authors of [56] already pointed out, these quantities are not directly comparable. The major point is the fact that only two isomers of ISO2 are able to undergo 1,6-H shifts, whereas all six isomers can react with HO_2 . Because mass spectrometry is not able to distinguish between iso-

mers, the rate coefficient k_{isom} in [56] had to be defined by $d[\text{HPALD}]/dt = k_{\text{isom}}[\text{ISO}_2]$ with $[\text{HPALD}] = [\text{HPALD}(1,4)] + [\text{HPALD}(4,1)]$, and $[\text{ISO}_2]$ being the overall concentration of all ISO2 isomers. Note that, in analogy to reactions (R4a) and (R4b), which produce HPALD(1,4), the isomer HPALD(4,1) is produced in reactions (R7a) and (R7b):



In [25], relative branching fractions for $\text{C}_5\text{H}_8 + \text{OH}$ of $[1\text{-ISOH}]/[2\text{-ISOH}]/[3\text{-ISOH}]/[4\text{-ISOH}] \sim 0.60/0.05/0.05/0.30$ were given. If one assumes a rapid equilibration between a given ISOH species and the respective ISO2 isomer(s) (cf., e.g., reactions (R2a) and (R2b)), one can estimate relative concentrations on the basis of these branching fractions. From relative energies and partition functions in the rigid rotor/harmonic oscillator approximation, we obtain $[\text{Z-1,4-ISO}_2]/[\text{ISO}_2] \sim 0.01$ and $[\text{Z-4,1-ISO}_2]/[\text{ISO}_2] \sim 0.005$. The energies and molecular data used are given in the Supporting Information. By assuming similar rate coefficients $k_{4a} \sim k_{7a}$, it follows $k_{4a} \sim 70 \times k_{\text{isom}}$, which explains a considerable part of the pretended discrepancy between the experimental and calculated results. A factor of 40, however, remains.

In [56], also the isotope effect was measured by using perdeuterated isoprene, C_5D_8 , as a reactant. The authors found values of $k_{\text{isom}}/k_{\text{isom}}(\text{d8}) \sim 14$ with a slight negative temperature dependence in the range $T = 295\text{--}318 \text{ K}$. We also calculated the isotope effect for the isomerization of Z-1,4-ISO2, using the IRC curve given by Nguyen et al. [6] and a recalculated curve for the deuterated species. We obtained values of $k_{4a}/k_{4a}(\text{d8})$ between 25 and 18 at temperatures between 318 and 295 K, that is, a factor of ~ 1.5 higher than the observed value. This difference seems tolerable in view of the approximate nature of the one-dimensional tunneling correction and the uncertainties in the vibrational wave numbers. We note that the contribution to the isotope effect from the difference of the threshold energies lies between 7.5 at $T = 295 \text{ K}$ and 6.3 at $T = 318 \text{ K}$, whereas tunneling causes an additional factor of 3.3 and 2.9, respectively. Nguyen et al. [6] calculated

isotope effects of $k_{4a}/k_{4a}(d8) \sim 37$ at $T = 295$ K and ~ 26 at $T = 318$ K, which is a factor of 1.5 higher than our values and a factor of 2 above the experimental results. On an absolute scale, our calculated rate coefficients k_{4a} and $k_{4a}(d8)$ are increased due to tunneling by a factor of 26 and 8, respectively ($T = 300$ K).

As a final note, we mention that an uncertainty of $\pm 50\%$ was assigned to the ratio k_{isom}/k_6 in [56]. The absolute value of k_{isom} depends on the accuracy of k_6 , for which the available database is rather sparse. To the best of our knowledge, there is only one experimental value. Boyd et al. [57] report $k_6 = 1.7 \times 10^{-11} \text{ cm}^3 \text{ s}^{-1}$ at $T = 298$ K and atmospheric pressure with an estimated error of 20%. The temperature dependence of k_6 adopted in [56] relies on a simple empirical structure–reactivity relationship [58]. For 298 K, the resulting rate coefficient is in good agreement with the value determined by Boyd et al. [57], but temperature-dependent measurements of k_6 would improve the situation.

CONCLUSIONS

In the present work, the influence of (double) chemical activation on consecutive complex-forming bimolecular reactions under steady-state conditions was examined. To this end, two master equations for chemical activation were coupled. The general approach was described, and the method was applied to intermediates in the OH-initiated atmospheric oxidation of isoprene. Emphasis was put on the unimolecular 1,5- and 1,6-H shift reactions of the ISO2 radicals. The following conclusions can be drawn:

1. In the reaction sequence isoprene + OH $\rightarrow$ ISOH and ISOH + O₂ $\rightarrow$ ISO2, with the latter step assumed to be irreversible, the energy distribution of ISOH was shown to be significantly different from a thermal distribution due to the rapid consecutive reaction with O₂. The back dissociation of ISOH into isoprene + OH can be neglected.
2. As a consequence, the energy distribution of ISO2 exhibits a very small fraction of residual “double activation.” This leads to increased rate coefficients compared to thermal activation. However, the branching ratios between forward and backward reactions are nearly unaffected. The back dissociation into ISOH + O₂ remains dominant.
3. Accordingly, the irreversibility assumption for the reaction ISOH + O₂ $\rightleftharpoons$ ISO2 has to be abandoned. Moreover, the timescale of equilibra-

tion is fast compared to the consecutive isomerization reactions (R3a) and (R4a). Therefore, a “double activation” mechanism is not operative, because ISOH is rapidly thermalized under atmospheric conditions. It was shown that a single chemical activation mechanism is also not relevant, because, in this case, ISO2 would be quickly stabilized.

4. High-pressure rate coefficients of the 1,5-H shift reaction of 1,2-ISO2 and the 1,6-H shift reaction of Z-1,4-ISO2 at $T = 298$ K were calculated ($k_{3a}^\infty = 1.5 \times 10^{-3} \text{ s}^{-1}$ and $k_{4a}^\infty = 6.5 \text{ s}^{-1}$, respectively). These are the relevant quantities to be used in the atmospheric oxidation mechanism of isoprene, even though the corresponding thermal reactions would be in the falloff regime. The calculated values are in reasonable agreement with data previously reported by Peeters et al. [5,6] and da Silva et al. [7].
5. Differences between calculated and experimentally determined rate coefficients are discussed in terms of isomer abundance and tunneling corrections. Whereas the isotope effect is satisfactorily reproduced, a considerable discrepancy (one order of magnitude) in the absolute values of the rate coefficients remains. Whether this is due to uncertainties in the calculated threshold energies or in the relative rate experiments remains an open question.

We thank Dr. T. Berndt for stimulating discussions.

BIBLIOGRAPHY

1. Atkinson, R.; Arey, J. *Chem Rev* 2003, 103, 4605–4638.
2. Lelieveld, J.; Butler, T. M.; Crowley, J. N.; Dillon, T. J.; Fischer, H.; Ganzeveld, L.; Harder, H.; Lawrence, M. G.; Martinez, M.; Taraborrelli, D.; Williams, J. *Nature* 2008, 452, 737–740.
3. Hofzumahaus, A.; Rohrer, F.; Lu, K.; Bohn, B.; Brauers, T.; Chang, C.-C.; Fuchs, H.; Holland, F.; Kita, K.; Kondo, Y.; Li, X.; Lou, S.; Shao, M.; Zeng, L.; Wahner, A.; Zhang, Y. *Science* 2009, 324, 1702–1704.
4. Lei, W.; Zhang, R.; McGivern, W. S.; Derecskei-Kovacs, A.; North, S. W. *J Phys Chem A* 2001, 105, 471–477.
5. Peeters, J.; Nguyen, T. L.; Vereecken, L. *Phys Chem Chem Phys* 2009, 11, 5935–5939.
6. Nguyen, T. L.; Vereecken, L.; Peeters, J. *ChemPhysChem* 2010, 11, 3996–4001.
7. da Silva, G.; Graham, C.; Wang, Z.-F. *Environ Sci Technol* 2010, 44, 250–256.
8. Peeters, J.; Müller, J.-F. *Phys Chem Chem Phys* 2010, 12, 14227–14235.

9. Stavrakou, T.; Peeters, J.; Müller, J.-F. *Atmos Chem Phys* 2010, 10, 9863–9878.
10. Archibald, A. T.; Cooke, M. C.; Utembe, S. R.; Shallcross, D. E.; Derwent, R. G.; Jenkin, M. E. *Atmos Chem Phys* 2010, 10, 8097–8118.
11. Gilbert, R. G.; Smith, S. C. *Theory of Unimolecular and Recombination Reactions*; Blackwell: Oxford, UK, 1990.
12. Holbrook, K. A.; Pilling, M. J.; Robertson, S. H. *Unimolecular Reactions*, 2nd ed.; Wiley: Chichester, UK, 1996.
13. Forst, W. *Unimolecular Reactions: A Concise Introduction*; Cambridge University Press: Cambridge, UK, 2003.
14. da Silva, G. *J Phys Chem A* 2010, 114, 6861–6869.
15. Sheng, C. Y.; Bozzelli, J. W.; Dean, A. M.; Chang, A. Y. *J Phys Chem A* 2002, 106, 7276–7293.
16. Hou, H.; Li, A.; Hu, H.; Li, Y.; Li, H.; Wang, B. *J Chem Phys* 2005, 122, 224304.
17. Maranzana, A.; Barker, J. R.; Tonachini, G. *Phys Chem Chem Phys* 2007, 9, 4129–4141.
18. da Silva, G. *Phys Chem Chem Phys* 2010, 12, 6698–6705.
19. da Silva, G.; Bozzelli, J. W.; Liang, L.; Farrell, J. T. *J Phys Chem A* 2009, 113, 8923–8933.
20. Glowacki, D. R.; Lockhart, J.; Blitz, M. A.; Klippenstein, S. J.; Pilling, M. J.; Robertson, S. H.; Seakins, P. W. *Science* 2012, 337, 1066–1069.
21. Asatryan, R.; da Silva, G.; Bozzelli, J. W. *J Phys Chem A* 2010, 114, 8302–8311.
22. da Silva, G. *J Phys Chem A* 2012, 116, 5317–5324.
23. Dibble, T. S. *J Phys Chem A* 2002, 106, 6643–6650.
24. Park, J.; Jongsma, C. G.; Zhang, R.; North, S. W. *J Phys Chem A* 2004, 108, 10688–10697.
25. Peeters, J.; Boullart, W.; Pultau, V.; Vandenberg, S.; Vereecken, L. *J Phys Chem A* 2007, 111, 1618–1631.
26. Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Montgomery, J. A., Jr.; Vreven, T.; Kudin, K. N.; Burant, J. C.; Millam, J. M.; Iyengar, S. S.; Tomasi, J.; Barone, V.; Mennucci, B.; Cossi, M.; Scalmani, G.; Rega, N.; Petersson, G. A.; Nakatsuji, H.; Hada, M.; Ehara, M.; Toyota, K.; Fukuda, R.; Hasegawa, J.; Ishida, M.; Nakajima, T.; Honda, Y.; Kitao, O.; Nakai, H.; Klene, M.; Li, X.; Knox, J. E.; Hratchian, H. P.; Cross, J. B.; Bakken, V.; Adamo, C.; Jaramillo, J.; Gomperts, R.; Stratmann, R. E.; Yazyev, O.; Austin, A. J.; Cammi, R.; Pomelli, C.; Ochterski, J. W.; Ayala, P. Y.; Morokuma, K.; Voth, G. A.; Salvador, P.; Dannenberg, J. J.; Zakrzewski, V. G.; Dapprich, S.; Daniels, A. D.; Strain, M. C.; Farkas, O.; Malick, D. K.; Rabuck, A. D.; Raghavachari, K.; Foresman, J. B.; Ortiz, J. V.; Cui, Q.; Baboul, A. G.; Clifford, S.; Cioslowski, J.; Stefanov, B. B.; Liu, G.; Liashenko, A.; Piskorz, P.; Komaromi, I.; Martin, R. L.; Fox, D. J.; Keith, T.; Al-Laham, M. A.; Peng, C. Y.; Nanayakkara, A.; Challacombe, M.; Gill, P. M. W.; Johnson, B.; Chen, W.; Wong, M. W.; Gonzalez, C.; Pople, J. A. *Gaussian 03, Revision C.02*; Gaussian: Wallingford, CT, 2004.
27. Olzmann, M.; Gebhardt, J.; Scherzer, K. *Int J Chem Kinet* 1991, 23, 825–835.
28. Olzmann, M. *Phys Chem Chem Phys* 2002, 4, 3614–3618.
29. González-García, N.; Olzmann, M. *Phys Chem Chem Phys* 2010, 12, 12290–12298.
30. Caralp, S.; Forst, W.; Bergeat, A. *Phys Chem Chem Phys* 2008, 10, 5746–5753.
31. Schranz, H.; Nordholm, S. *Chem Phys* 1984, 87, 163–177.
32. Snider, N. *J Chem Phys* 1984, 80, 1885–1893.
33. Pritchard, H. O.; Vatsya, S. R. *J Phys Chem* 1984, 88, 5816–5819.
34. Forst, W. *Theory of Unimolecular Reactions*; Academic Press: New York, 1973.
35. Oref, I.; Tardy, D. C. *Chem Rev* 1990, 90, 1407–1445.
36. Simmrock, K. H.; Janowsky, R.; Ohnsorge, A. *Critical Data of Pure Substances*; DECHEMA: Frankfurt/Main, Germany, 1996.
37. Stiel, L. I.; Thodos, G. *AIChE J* 1964, 10, 266–269.
38. Reid, R. C.; Prausnitz, J. M.; Poling, B. E. *The Properties of Gases and Liquids*, 4th ed.; McGraw-Hill: Boston, MA, 1987.
39. Smith, B. T.; Boyle, J. M.; Garbow, B. S.; Ikebe, Y.; Klema, V. C.; Moler, C. B. *Matrix Eigensystems Routines—EISPACK Guide*; Goos, G.; Hartmanis, J., Eds.; Lecture Notes in Computer Science; Vol. 6, Springer: Berlin, 1974.
40. Press, W. H.; Teukolsky, S. A.; Vetterling, W. T.; Flannery, B. P. *Numerical Recipes in Fortran: The Art of Scientific Computing*, 2nd ed.; Cambridge University Press: Cambridge, UK, 1992.
41. Olzmann, M. In *Cleaner Combustion: Developing Detailed Chemical Kinetic Models*; Battin-Leclerc, F.; Simmie, J. M.; Blurock, E., Eds.; Springer: London, 2013; Chap. 21.
42. Troe, J. *J Chem Phys* 1983, 79, 6017–6029.
43. Olzmann, M. *Ber Bunsen-Ges Phys Chem* 1997, 101, 533–537.
44. Marcus, R. A.; Rice, O. K. *J Phys Colloid Chem* 1951, 55, 894–908.
45. Marcus, R. A. *J Chem Phys* 1952, 20, 359–364.
46. Miller, W. H. *J Am Chem Soc* 1979, 101, 6810–6814.
47. Garrett, B. C.; Truhlar, D. G. *J Phys Chem* 1979, 83, 2921–2926.
48. Wentzel, G. *Z Phys* 1926, 38, 518–529.
49. Kramers, H. A. *Z Phys* 1926, 39, 828–840.
50. Brillouin, L. *J Phys Radium* 1926, 7, 353–368.
51. Quack, M.; Troe, J. *Ber Bunsen-Ges Phys Chem* 1974, 78, 240–252.
52. Troe, J. *J Chem Phys* 1981, 75, 226–237.
53. Atkinson, R.; Baulch, D. L.; Cox, R. A.; Crowley, J. N.; Hampson, R. F.; Hynes, R. G.; Jenkin, M. E.; Rossi, M. J.; Troe, J. *Atmos Chem Phys* 2006, 6, 3625–4055.
54. Merrick, J. P.; Moran, D.; Radom, L. *J Phys Chem A* 2007, 111, 11683–11700.

55. Hoyer mann, K.; Maarfeld, S.; Nacke, F.; Nothdurft, J.; Olzmann, M.; Wehmeyer, J.; Welz, O.; Zeuch, T. *Phys Chem Chem Phys* 2010, 12, 8953–8967.
56. Crounse, J. D.; Paulot, F.; Kjaergaard, H. G.; Wennberg, P. O. *Phys Chem Chem Phys* 2011, 13, 13607–13613.
57. Boyd, A. A.; Flaud, P. M.; Daugey, N.; Lesclaux, R. *J Phys Chem A* 2003, 107, 818–821.
58. Saunders, S. M.; Jenkin, M. E.; Derwent, R. G.; Pilling, M. J. *Atmos Chem Phys* 2003, 3, 161–180.
59. Lei, W.; Derecskei-Kovacs, A.; Zhang, R. *J Chem Phys* 2000, 113, 5354–5360.
60. Allodi, M. A.; Kirschner, K. N.; Shields, G. C. *J Phys Chem A* 2008, 112, 7064–7071.
61. Greenwald, E. E.; North, S. W.; Georgievskii, Y.; Klippenstein, S. J. *J Phys Chem A* 2007, 111, 5582–5592.
62. Francisco-Marquez, M.; Alvarez-Idaboy, J. R.; Galano, A.; Vivier-Bunge, A. *Phys Chem Chem Phys* 2003, 5, 1392–1399.